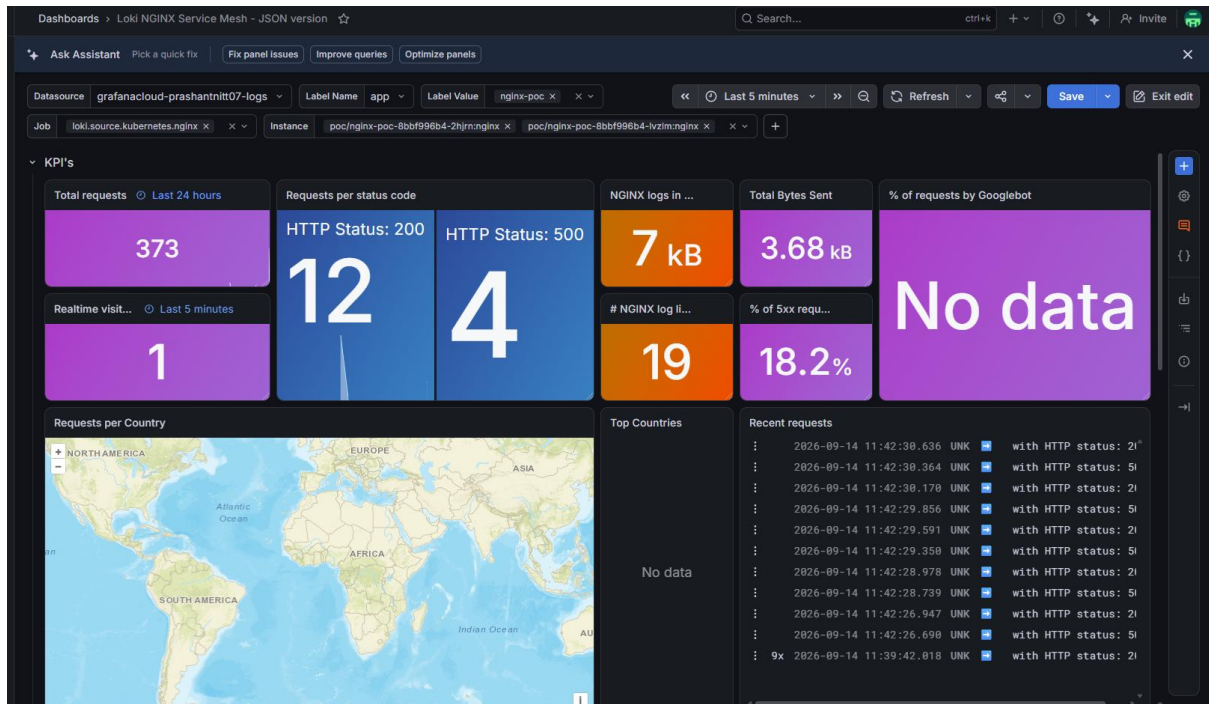


NGINX → Grafana Alloy → Grafana Cloud Loki → Grafana Cloud Dashboard

<https://grafana.com/grafana/dashboards/12559-loki-nginx-service-mesh-json-version/>



Update:

```
sudo apt update
```

```
sudo apt upgrade -y
```

Install utilities:

```
sudo apt install -y \
```

```
curl \
```

```
wget \
```

```
git \
```

```
jq \
```

```
stress-ng \
```

```
unzip
```

Install k3s

```
curl -sfL https://get.k3s.io | sh -
```

create ns:

```
00-namespace.yaml      01-nginx-config.yaml      02-nginx-deployment.yaml  07-nginx-service.yaml
root@ip-172-31-19-145:/home/ubuntu/nginx-grafana-cloud-poc# cat 00-namespace.yaml
apiVersion: v1
kind: Namespace
metadata:
  name: poc
root@ip-172-31-19-145:/home/ubuntu/nginx-grafana-cloud-poc#
root@ip-172-31-19-145:/home/ubuntu/nginx-grafana-cloud-poc#
```

Create 01-nginx-config.yaml

```

root@ip-172-31-19-145:/home/ubuntu/nginx-grafana-cloud-poc#
root@ip-172-31-19-145:/home/ubuntu/nginx-grafana-cloud-poc# cat 01-nginx-config.yaml
apiVersion: v1
kind: ConfigMap
metadata:
  name: nginx-config
  namespace: poc
data:
  nginx.conf: |
    worker_processes auto;

    events {
      worker_connections 1024;
    }

    http {

      log_format json_combined escape=json
      '{'
        '"time":'$time_iso8601','
        '"remote_addr":'$remote_addr','
        '"request":'$request','
        '"method":'$request_method','
        '"uri":'$uri','
        '"status":'$status','
        '"body_bytes_sent":'$body_bytes_sent','
        '"request_time":'$request_time','
        '"upstream_response_time":'$upstream_response_time','
        '"http_user_agent":'$http_user_agent','
        '"http_referer":'$http_referer'
      '}';

      access_log /dev/stdout json_combined;
      error_log /dev/stderr warn;

      server {

        listen 80;

        server_name _;

        location / {

          default_type application/json;

          return 200
            '{"message":"NGINX Grafana Cloud POC","status":"success"}';
        }

        location /health {
          default_type application/json;

          return 200
            '{"status":"healthy"}';
        }

        location /error {
          return 500;
        }

        location /notfound {
          return 404;
        }
      }
    }

```

Nginx deployment :

```

root@ip-172-31-19-145:/home/ubuntu/nginx-grafana-cloud-poc# cat 02-nginx-deployment.yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: nginx-poc
  namespace: poc
spec:
  replicas: 2
  selector:
    matchLabels:
      app: nginx-poc
  template:
    metadata:
      labels:
        app: nginx-poc
    spec:
      containers:
        - name: nginx
          image: nginx:1.27-alpine
          ports:
            - containerPort: 80
          volumeMounts:
            - name: nginx-config
              mountPath: /etc/nginx/nginx.conf
              subPath: nginx.conf
      volumes:
        - name: nginx-config
          configMap:
            name: nginx-config
root@ip-172-31-19-145:/home/ubuntu/nginx-grafana-cloud-poc# █

```

Nginx service:

```

root@ip-172-31-19-145:/home/ubuntu/nginx-grafana-cloud-poc# cat 07-nginx-service.yaml
apiVersion: v1
kind: Service
metadata:
  name: nginx-poc
  namespace: poc
spec:
  selector:
    app: nginx-poc
  ports:
    - name: http
      port: 80
      targetPort: 80
      nodePort: 30900
  type: NodePort
root@ip-172-31-19-145:/home/ubuntu/nginx-grafana-cloud-poc# █

```

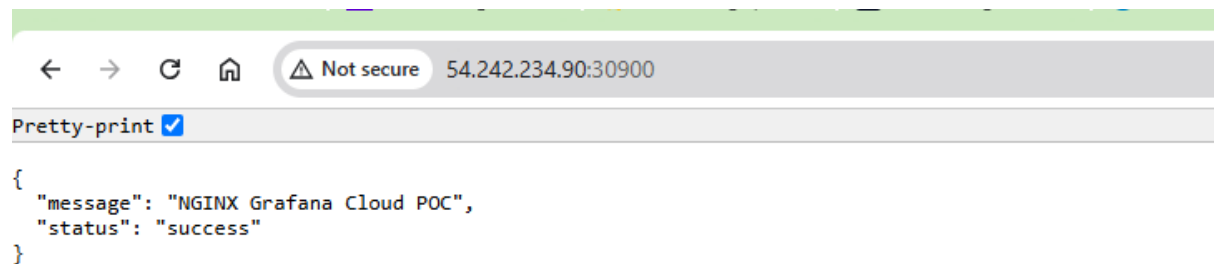
Nginx validation:

```
root@ip-172-31-19-145:/home/ubuntu/nginx-grafana-cloud-poc# kubectl get all -n poc
NAME                                READY   STATUS    RESTARTS   AGE
pod/nginx-poc-8bbf996b4-2hjrn      1/1     Running   0           7m28s
pod/nginx-poc-8bbf996b4-lvzlm      1/1     Running   0           7m28s

NAME                                TYPE           CLUSTER-IP   EXTERNAL-IP   PORT(S)        AGE
service/nginx-poc                  NodePort       10.43.170.160 <none>        80:30900/TCP   6m26s

NAME                                READY   UP-TO-DATE   AVAILABLE   AGE
deployment.apps/nginx-poc          2/2     2             2           7m28s

NAME                                DESIRED   CURRENT   READY   AGE
replicaset.apps/nginx-poc-8bbf996b4 2         2         2       7m28s
root@ip-172-31-19-145:/home/ubuntu/nginx-grafana-cloud-poc#
```



The screenshot shows a web browser window with the address bar displaying "54.242.234.90:30900". The page content shows a JSON response: {"message": "NGINX Grafana Cloud POC", "status": "success"}.

Grafana Cloud Secret

```
kubectl create secret generic grafana-cloud-loki \
-n poc \
--from-literal=username='YOUR_GRAFANA_CLOUD_LOKI_USERNAME' \
--from-literal=password='YOUR_GRAFANA_CLOUD_API_TOKEN'
```

Rbac role:

```

root@ip-172-31-19-145:/home/ubuntu/alloy# cat 03-alloy-rbac.yaml
apiVersion: v1
kind: ServiceAccount
metadata:
  name: alloy
  namespace: poc
---
apiVersion: rbac.authorization.k8s.io/v1
kind: ClusterRole
metadata:
  name: alloy
rules:
  - apiGroups:
    - ""
    resources:
    - pods
    - pods/log
    - namespaces
    verbs:
    - get
    - list
    - watch
---
apiVersion: rbac.authorization.k8s.io/v1
kind: ClusterRoleBinding
metadata:
  name: alloy
roleRef:
  apiGroup: rbac.authorization.k8s.io
  kind: ClusterRole
  name: alloy
subjects:
  - kind: ServiceAccount
    name: alloy
    namespace: poc
root@ip-172-31-19-145:/home/ubuntu/alloy# kubectl apply -f 03-alloy-rbac.yaml
serviceaccount/alloy created
clusterrole.rbac.authorization.k8s.io/alloy created
clusterrolebinding.rbac.authorization.k8s.io/alloy created

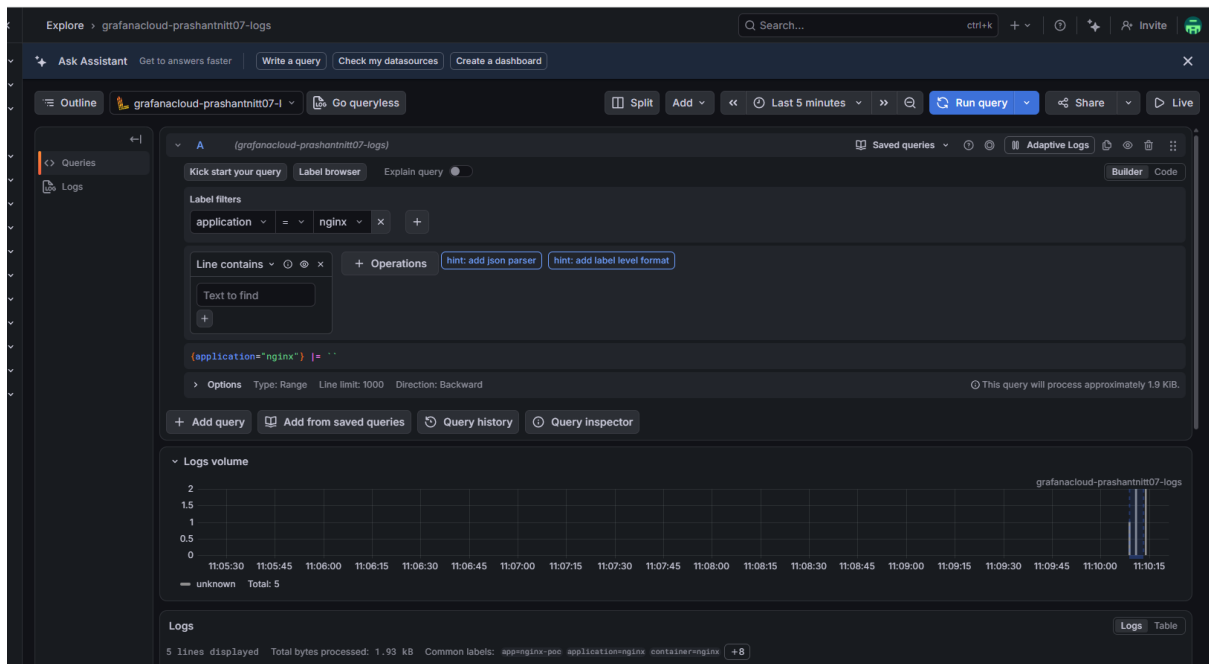
```

04-alloy-config.yaml

This is the most important file.

It:

1. Discovers Kubernetes Pods
2. Collects Pod logs
3. Keeps useful Kubernetes labels
4. Parses NGINX JSON
5. Sends logs to Grafana Cloud Loki



Generate some traffic:

```
for i in {1..100}; do
```

```
  curl -s http://NODE-IP:NODEPORT/ > /dev/null
```

```
done
```

Generate 404:

```
for i in {1..30}; do
```

```
  curl -s http://NODE-IP:NODEPORT/notfound > /dev/null
```

```
done
```

Generate 500:

```
for i in {1..20}; do
```

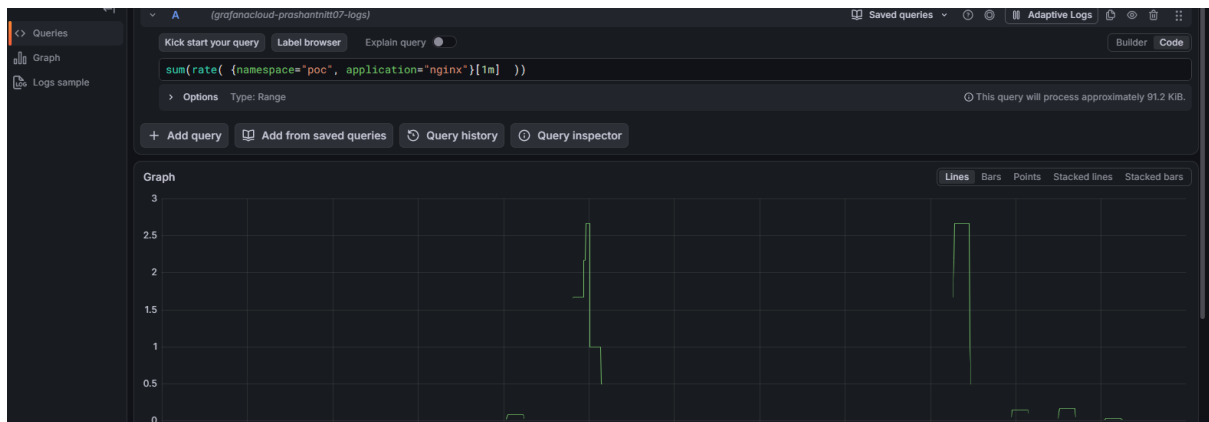
```
  curl -s http://NODE-IP:NODEPORT/error > /dev/null
```

```
done
```

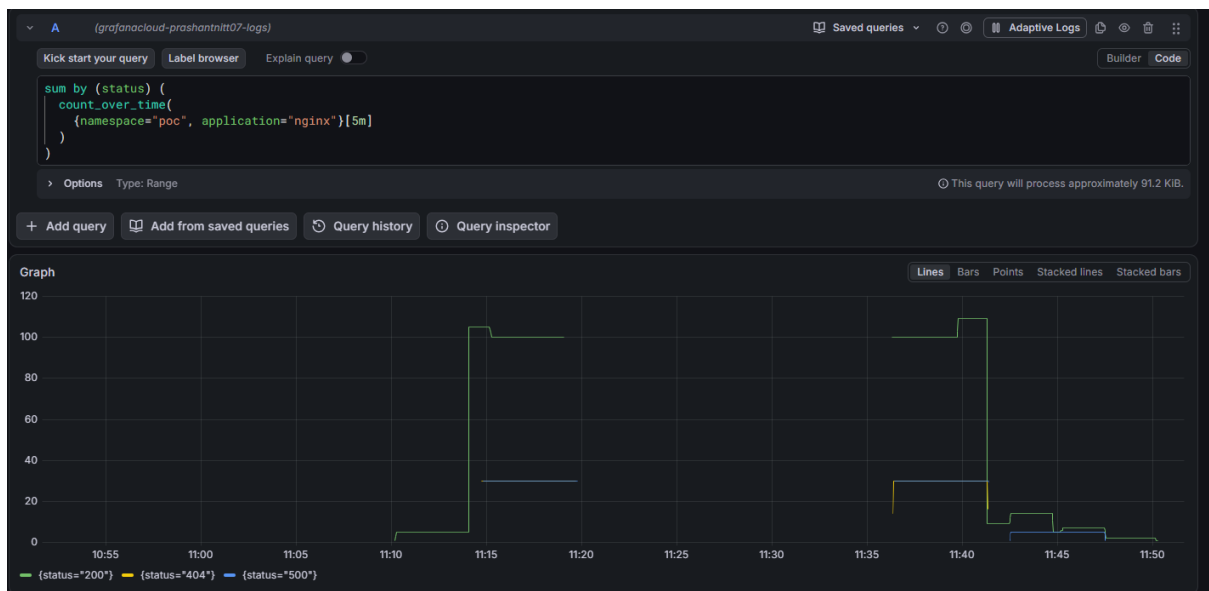
Grafana dashboard import -> 12559

LogQL :
Requests/sec

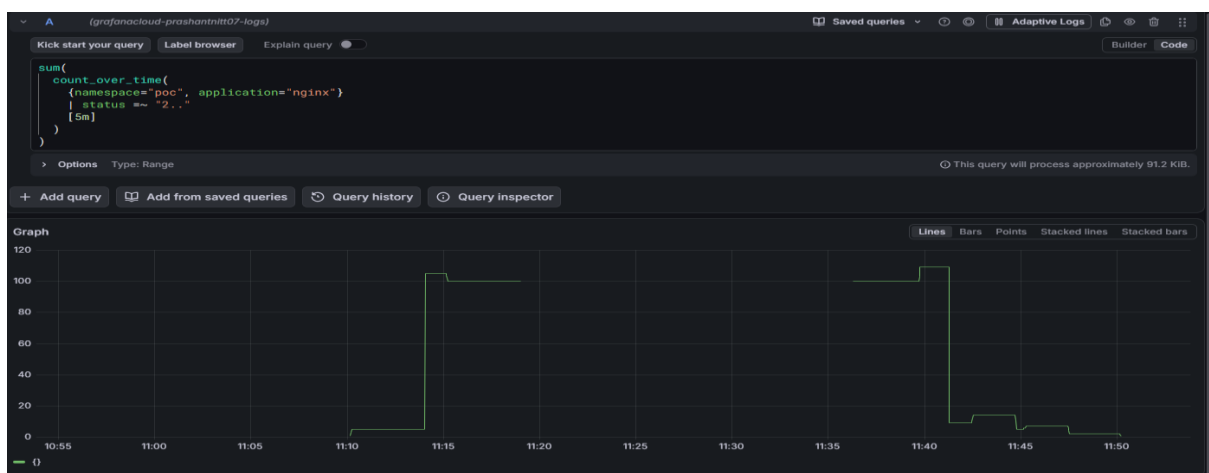
```
sum(rate( {namespace="poc", application="nginx"}[1m] ))
```



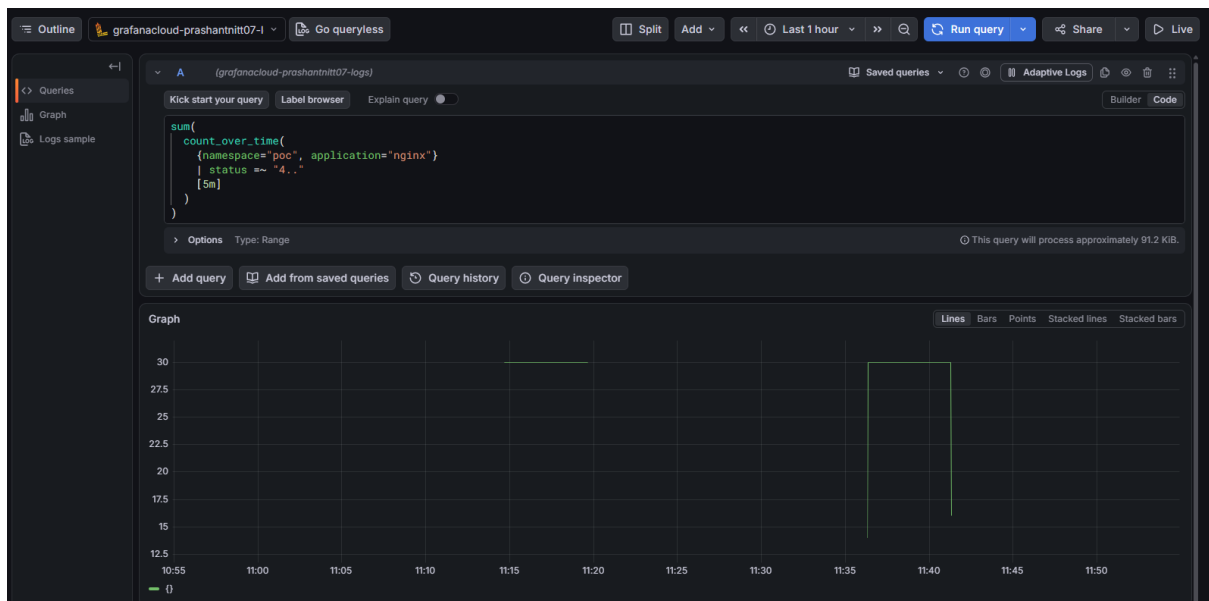
Status code



2xx



4xx



5xx

